

BARRIE-ORO, ON, Canada

WMO: 713140

Lat: **44.484N**Lon: **79.546W**Elev: **289**StdP: **97.90**Time Zone: **-5.00 (NAE)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.3	-20.7	-28.1	0.3	-24.0	-24.5	0.4	-20.2	10.9	-4.1	9.7	-5.9	2.2	0	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.4	21.4	27.7	20.6	26.3	19.7	22.7	27.4	21.7	26.2	20.9	24.8	4.0	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.4	25.2	20.2	15.5	24.3	19.4	14.7	23.5	68.5	27.6	65.0	26.2	61.8	24.8	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.5	8.4	7.4	-29.9	31.8	3.0	1.3	-32.1	32.8	-33.8	33.5	-35.5	34.2	-37.6	35.2	
(5)				-30.1	24.6	2.9	0.9	-32.2	25.2	-33.9	25.7	-35.6	26.2	-37.7	26.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.3	-8.3	-7.7	-2.5	5.2	12.2	16.8	19.6	18.5	14.9	8.5	1.9	-4.2
	DBStd	11.02	7.20	6.00	6.31	4.73	4.67	3.63	2.96	2.94	3.96	4.42	5.00	5.70
	HDD10.0	2410	567	494	391	157	29	1	0	0	5	83	243	439
	HDD18.3	4529	826	728	647	396	199	71	21	33	116	304	492	697
	CDD10.0	1069	0	0	2	12	97	205	298	264	153	37	2	0
	CDD18.3	148	0	0	0	0	8	25	60	39	14	1	0	0
	CDH23.3	1285	0	0	1	5	102	225	503	312	129	9	0	0
	CDH26.7	261	0	0	0	0	19	48	112	58	23	1	0	0
Wind	WSAvg	3.3	3.4	3.6	3.8	4.1	3.4	3.1	2.8	2.7	2.7	3.3	3.4	3.5
Precipitation	PrecAvg	871	65	55	58	67	73	85	80	79	82	77	76	75
	PrecMax	1114	106	112	156	145	141	206	120	141	140	139	143	116
	PrecMin	681	31	24	16	22	27	22	32	29	34	28	30	35
	PrecStd	93	22	24	29	29	26	41	27	27	25	29	29	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	8.8	17.9	23.7	28.9	30.2	31.5	30.8	29.5	24.6	16.9	10.5
		MCWB	8.2	6.2	13.2	14.9	20.0	22.3	23.0	22.1	21.7	18.1	12.5	9.3
	2%	DB	5.9	5.1	11.9	19.4	26.4	28.1	29.5	28.2	26.7	21.1	13.6	7.2
		MCWB	4.6	3.6	8.4	11.7	18.3	20.8	21.5	20.8	20.2	16.1	10.7	6.0
	5%	DB	3.2	2.4	8.8	16.4	23.6	26.0	27.7	26.4	24.6	18.3	11.4	4.8
		MCWB	2.0	0.9	5.9	10.6	16.4	19.5	20.7	20.1	19.1	14.6	8.9	3.6
	10%	DB	1.2	0.8	5.9	13.6	21.1	23.9	26.1	24.9	22.4	15.9	9.2	2.8
		MCWB	0.2	-0.3	3.6	8.4	14.8	17.9	19.7	19.1	17.9	13.1	7.3	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	6.9	14.1	15.5	21.2	23.4	24.2	23.6	22.7	19.8	13.3	9.6
		MCDB	8.8	8.2	17.2	20.3	27.1	28.4	29.6	27.9	27.2	22.8	15.5	10.5
	2%	WB	4.7	3.8	9.2	13.4	19.5	21.9	22.7	22.1	21.4	17.0	11.0	6.1
		MCDB	5.5	5.1	11.2	18.3	24.6	27.1	27.4	26.5	25.4	19.8	13.3	7.1
	5%	WB	2.1	1.2	6.1	11.2	17.9	20.5	21.8	21.1	20.1	15.1	9.1	3.9
		MCDB	3.1	2.2	8.3	15.3	22.4	24.5	26.1	24.9	23.2	17.8	11.0	4.8
	10%	WB	0.3	-0.3	3.6	8.9	16.0	19.1	20.9	20.1	18.6	13.5	7.5	1.8
		MCDB	1.2	0.8	5.6	13.4	19.5	22.5	24.8	23.6	21.6	15.6	9.3	2.7
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.2	9.4	10.4	12.0	11.0	11.2	11.1	11.3	9.1	8.2	7.0
		MCDBR	8.2	10.3	12.8	16.2	15.2	13.4	12.9	13.1	13.8	13.5	11.5	8.1
		MCWBR	7.7	9.3	9.3	10.1	8.4	7.8	6.8	7.1	8.1	8.8	9.2	7.5
	5% WB	MCDBR	8.1	9.8	11.6	14.3	13.2	12.0	11.1	10.9	11.5	11.4	10.5	8.0
		MCWBR	7.8	9.2	9.0	10.0	8.0	7.6	6.5	6.7	7.4	8.3	8.8	7.6
Clear-Sky Solar Irradiance	taub	0.286	0.297	0.317	0.354	0.385	0.406	0.405	0.396	0.362	0.342	0.308	0.283	
	taud	2.384	2.340	2.410	2.364	2.326	2.287	2.308	2.323	2.415	2.458	2.502	2.453	
	Ebn at Noon	837	892	921	908	886	865	860	855	860	830	809	809	
	Edh at Noon	75	94	101	116	124	130	125	119	100	84	68	64	
All-Sky Solar Radiation	RadAvg	1.40	2.25	3.57	4.49	5.39	5.87	5.95	5.13	4.04	2.35	1.40	1.01	
	RadStd	0.16	0.28	0.38	0.44	0.55	0.45	0.39	0.27	0.39	0.25	0.19	0.12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page